

```

root_path = fileparts(mfilename("fullpath"));
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit400', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
%cuts_list = cuts2fit400.cutsS_list; % all available cuts
%cuts_list = {cuts2fit400.cut_b200_200sel}; % cuts in low ff direction around
<200>
cuts_list = cuts2fit400.cuts_dir001FFfit(1);
%cuts_list = cuts2fit400.cuts_dir001FFfit(2);

mi = maps_instrument(400,600, 'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
for i=1: numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
end

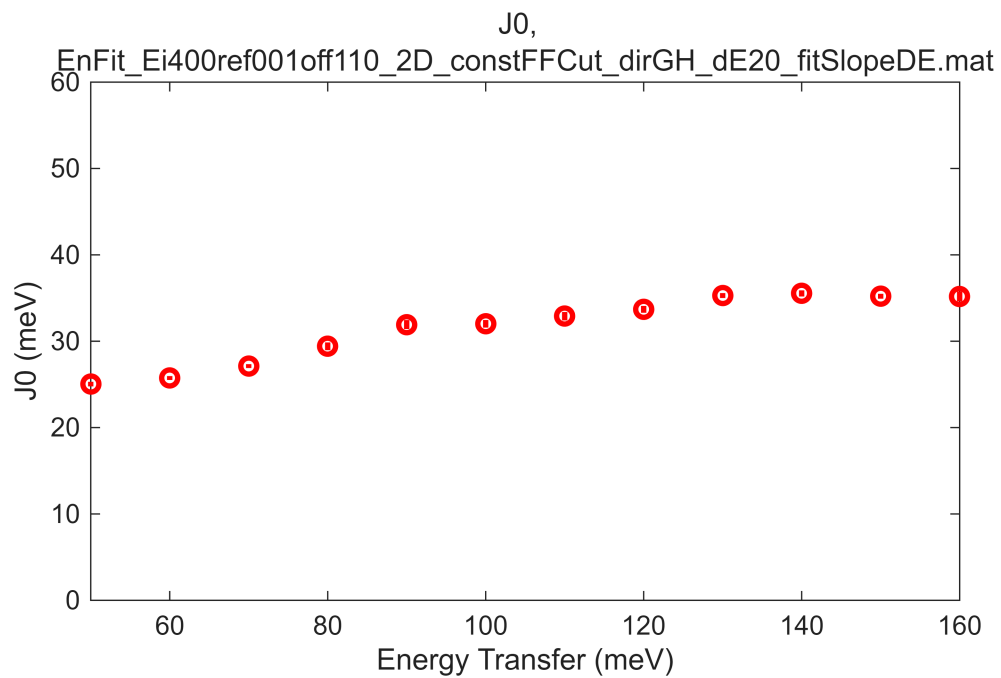
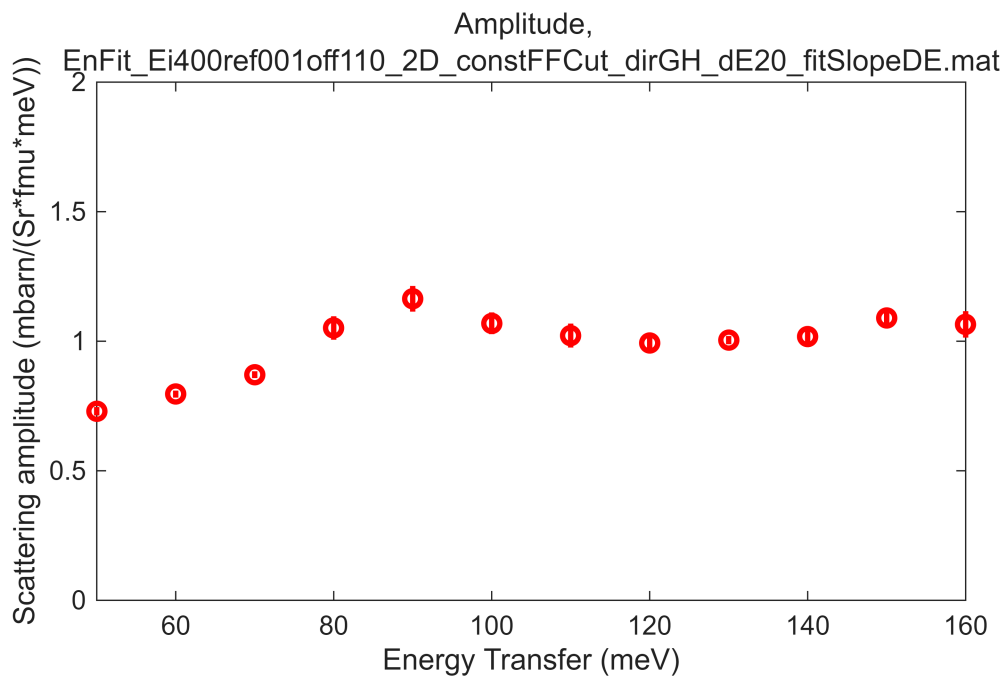
dir_name = "GH";

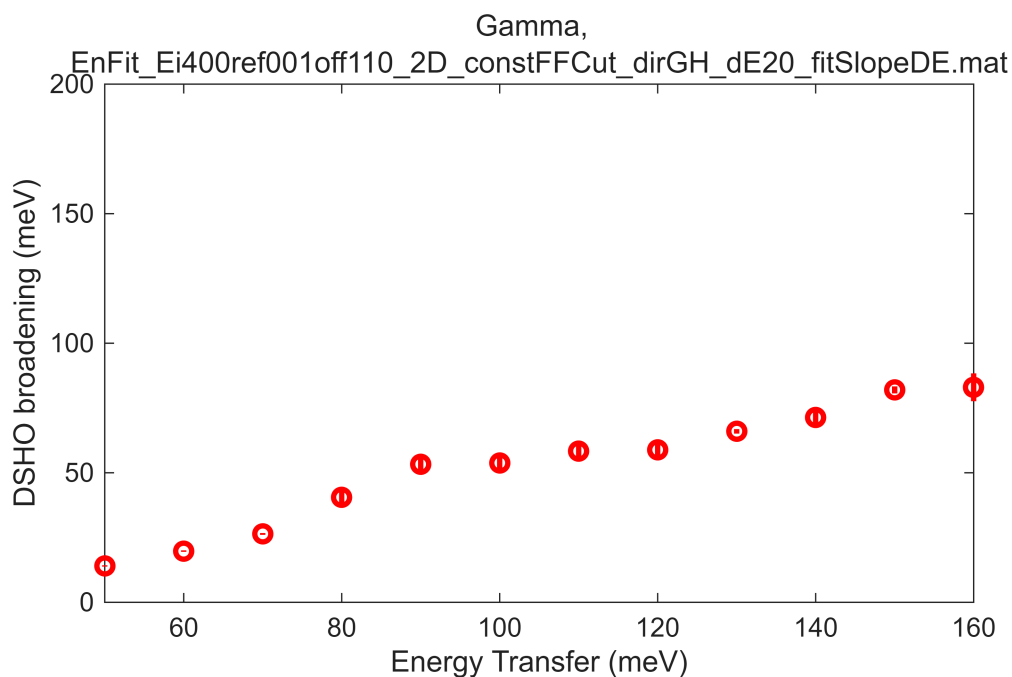
dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
%cut_en = 50:10:260;
cut_en = 50:10:160; %cuts2fit400.cuts_dir001FFfit(1);
%cut_en = 50:10:200; %cuts2fit400.cuts_dir001FFfit(2);

%cl= {cuts_list{3}};
%cl = {w2_test_cut};
% if isfield(cuts2fit400, 'bg_par_010off000')
%     if numel(cuts_list) == 3
%         bg_par =
{cuts2fit400.bg_par_010off000.p, cuts2fit400.bg_par_010off100.p, cuts2fit400.bg_par
_010off200.p};
%     else
%         bg_par = {cuts2fit400.bg_par_010off100.p};
%     end
%     cuts_list{end+1} = bg_par;
% end
if isfield(cuts2fit400, 'bg_par_010off200')
    if numel(cuts_list) == 3
        bg_par =
{cuts2fit400.bg_par_010off000.p, cuts2fit400.bg_par_010off100.p, cuts2fit400.bg_par
_010off200.p};
    else
        bg_par = {cuts2fit400.bg_par_010off200.p};
    end
    cuts_list{end+1} = bg_par;
end
end

```

```
[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
cuts_list,'EnFit_Ei400ref001off110_2D_constFFCut',dir_name,cut_en,dE_step,half_dE
);
```





**** En = 50 ± 10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 137832 pixels -- processing data... -----> included 113180 pixels

Number of points in input object: 137832

Fraction of object selected: 100.00% (= 137832 points) ; Access speed: 1.70GB/sec

Fraction of object retained: 82.11% (= 113180 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 34.8%; Transf. time frac.: 53.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 137832 pixels -- processing data... -----> retained 113180 pixels

Number of points in input object: 137832

Fraction of object selected: 100.00% (= 137832 points) ; Access speed: 1.70GB/sec

Fraction of object retained: 82.11% (= 113180 points)

**** Total time in cut(sqw) : 0.0sec Including:

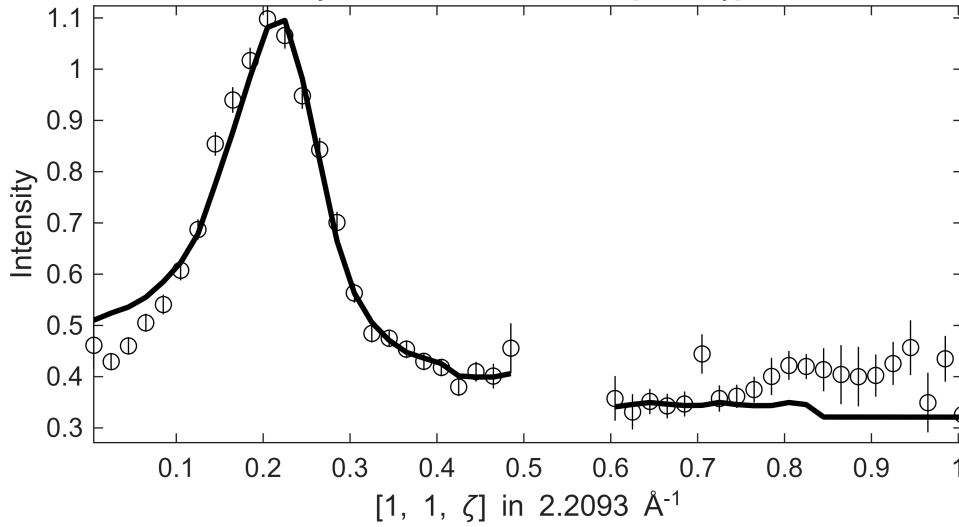
Data access time fraction: 16.2%; Transf. time frac.: 55.1%; Resulting pix preparation: 25.7%

Fe_ei401_align_sym3.sqw

S=0.73±0.016; J0= 25±0.27; gamma=14.0264±0.258334

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 40 ≤ E ≤ 60

ζ=-0.005:0.02:1.015 in [1, 1, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 137832 pixels -- processing data... -----> retained 113180 pixels

Number of points in input object: 137832

Fraction of object selected: 100.00% (= 137832 points) ; Access speed: 1.77GB/sec

Fraction of object retained: 82.11% (= 113180 points)

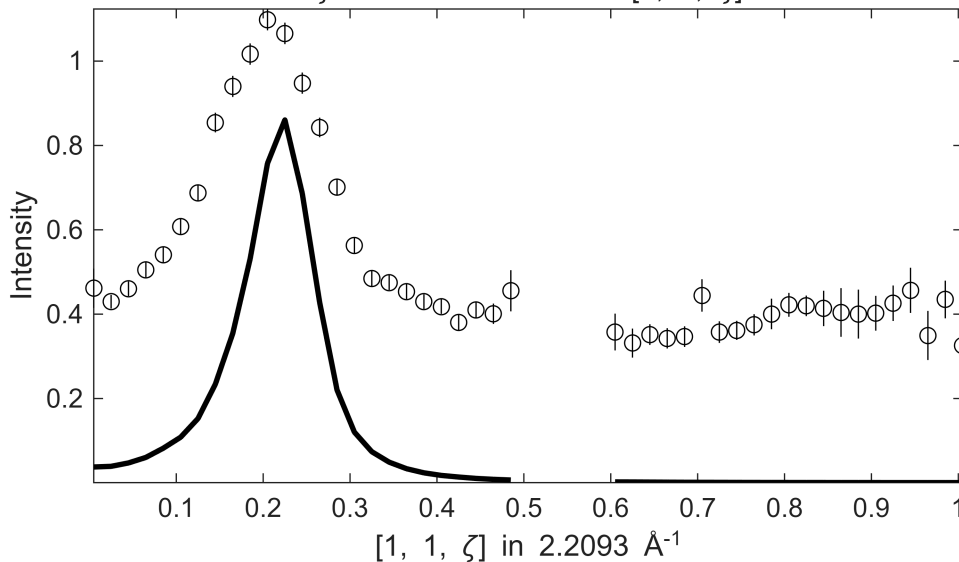
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 18.9%; Transf. time frac.: 49.8%; Resulting pix preparation: 31.2%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 40 ≤ E ≤ 60

ζ=-0.005:0.02:1.015 in [1, 1, ζ]



**** En = 60±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 185869 pixels -- processing data... -----> included 157263 pixels

Number of points in input object: 185869

Fraction of object selected: 100.00% (= 185869 points) ; Access speed: 2.02GB/sec

Fraction of object retained: 84.61% (= 157263 points)

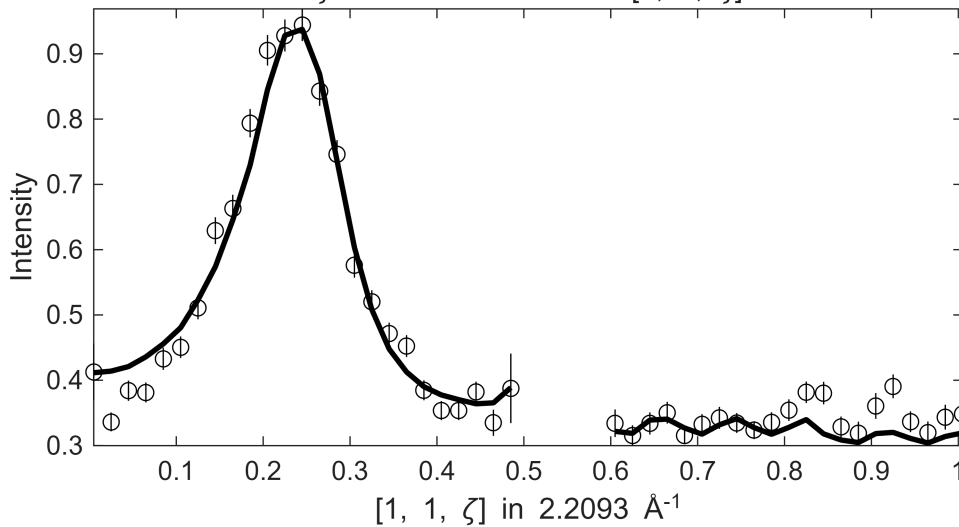
```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 45.9%; Transf. time frac.: 50.1%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 185866 pixels -- processing data... -----> retained 157260 pixels
-----
Number of points in input object: 185866
      Fraction of object selected: 100.00% (= 185866 points) ; Access speed: 1.96GB/sec
      Fraction of object retained: 84.61% (= 157260 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 27.5%; Transf. time frac.: 51.2%; Resulting pix preparation: 21.2%
-----

```

Fe_ei401_align_sym3.sqw
 $S = 0.8 \pm 0.035$; $J_0 = 26 \pm 0.32$; $\gamma = 19.8807 \pm 1.5422$
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $50 \leq E \leq 70$
 $\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 185869 pixels -- processing data... -----> retained 157263 pixels
-----
Number of points in input object: 185869
      Fraction of object selected: 100.00% (= 185869 points) ; Access speed: 1.94GB/sec
      Fraction of object retained: 84.61% (= 157263 points)

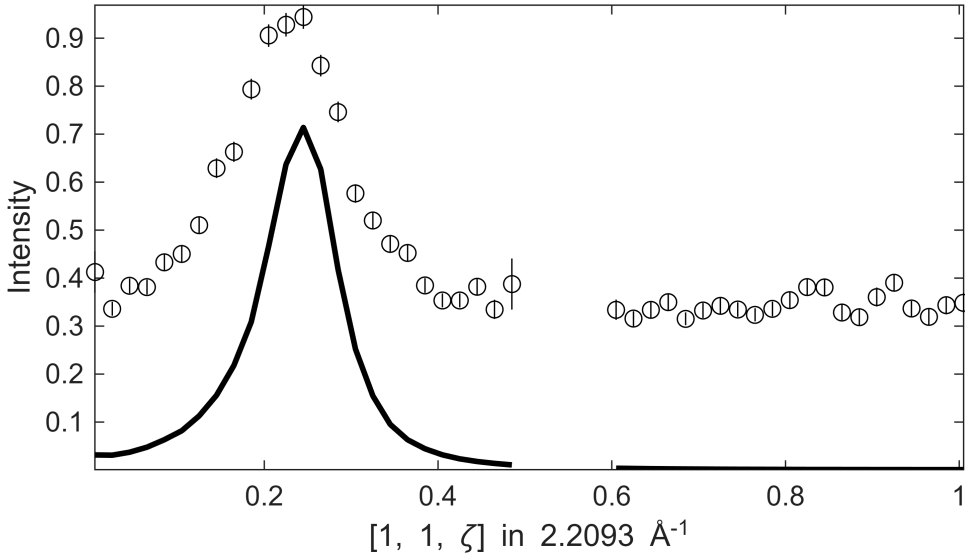
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 20.0%; Transf. time frac.: 54.8%; Resulting pix preparation: 25.1%
-----

```

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $50 \leq E \leq 100$ eV

$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



**** En = 70±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 218322 pixels -- processing data... -----> included 181783 pixels

Number of points in input object: 218322

Fraction of object selected: 100.00% (= 218322 points) ; Access speed: 1.87GB/sec

Fraction of object retained: 83.26% (= 181783 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 38.9%; Transf. time frac.: 58.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 218322 pixels -- processing data... -----> retained 181783 pixels

Number of points in input object: 218322

Fraction of object selected: 100.00% (= 218322 points) ; Access speed: 1.90GB/sec

Fraction of object retained: 83.26% (= 181783 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 23.1%; Transf. time frac.: 58.2%; Resulting pix preparation: 18.7%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 218322 pixels -- processing data... -----> retained 181783 pixels

Number of points in input object: 218322

Fraction of object selected: 100.00% (= 218322 points) ; Access speed: 1.71GB/sec

Fraction of object retained: 83.26% (= 181783 points)

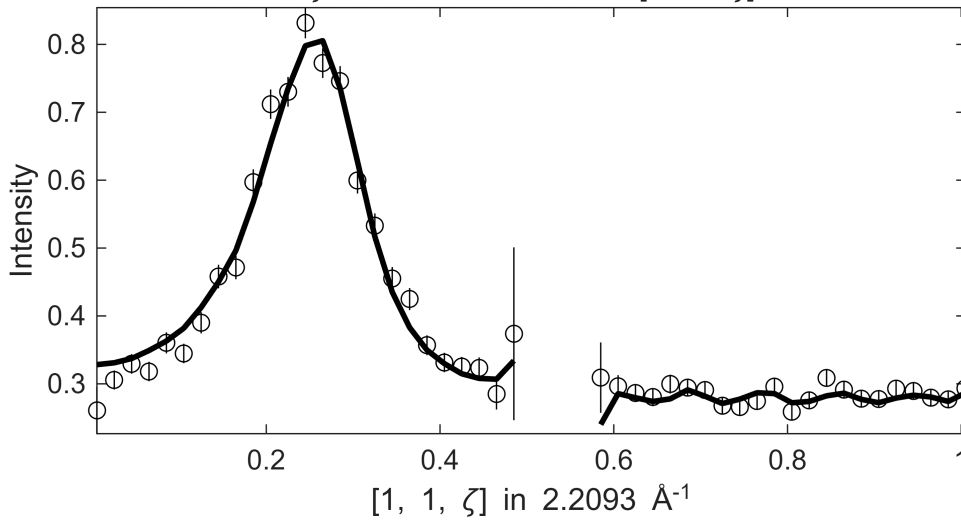
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 23.4%; Transf. time frac.: 58.4%; Resulting pix preparation: 18.2%

Fe_ei401_align_sym3.sqw

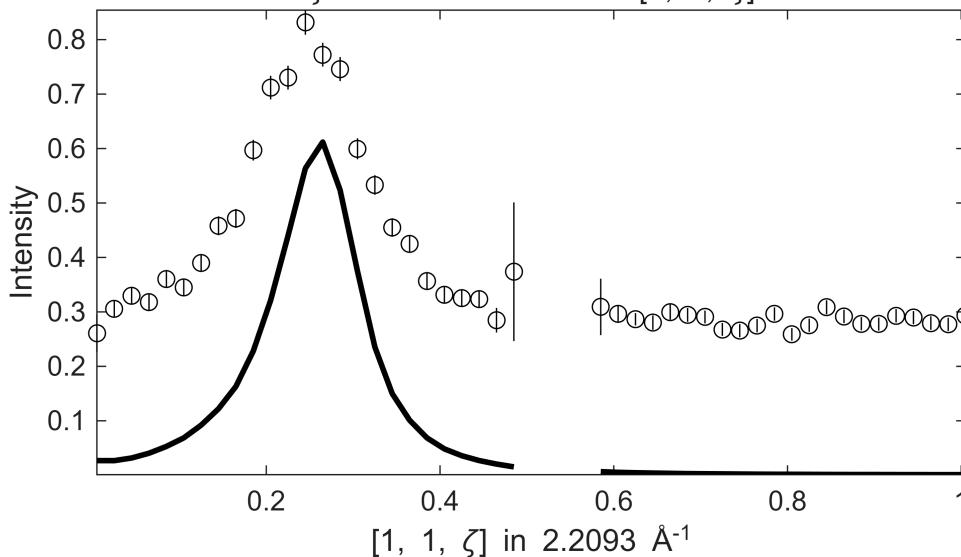
S=0.87±0.013; J0= 27±0.23; gamma=26.2271±0.359452

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 60 ≤ E ≤ 100 eV
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 60 ≤ E ≤ 100 eV
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



**** En = 80±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 226692 pixels -- processing data... -----> included 188732 pixels

Number of points in input object: 226692

Fraction of object selected: 100.00% (= 226692 points) ; Access speed: 2.03GB/sec

Fraction of object retained: 83.25% (= 188732 points)

*** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 46.5%; Transf. time frac.: 50.1%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; returning pixels - kept

*** Step 1 of 1; Read data for 226689 pixels -- processing data... -----> retained 188732 pixels

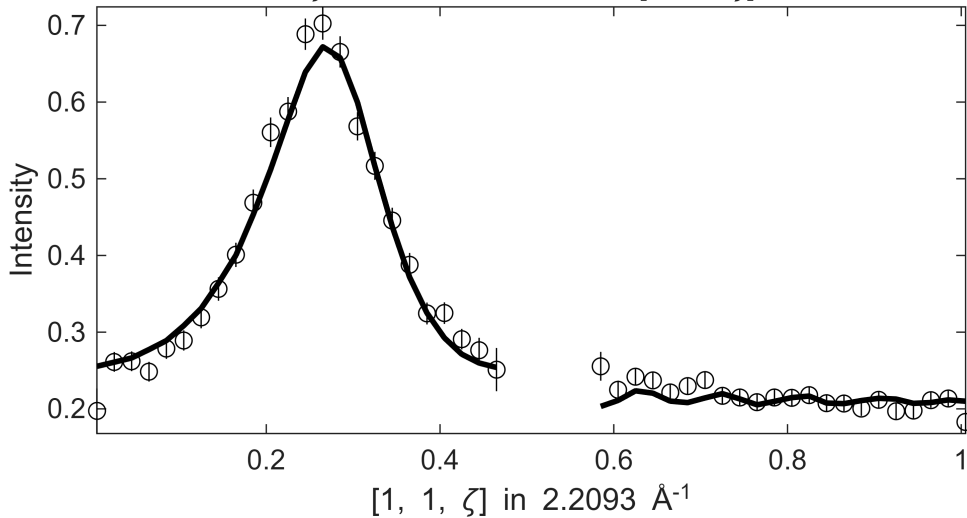
Number of points in input object: 226689

Fraction of object selected: 100.00% (= 226689 points) ; Access speed: 1.86GB/sec

Fraction of object retained: 83.26% (= 188732 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 30.1%; Transf. time frac.: 48.5%; Resulting pix preparation: 21.3%

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.015$; $J_0 = 29 \pm 0.24$; $\gamma = 40.5761 \pm 0.490747$
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $70 \leq E \leq 90$ eV
 $\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$

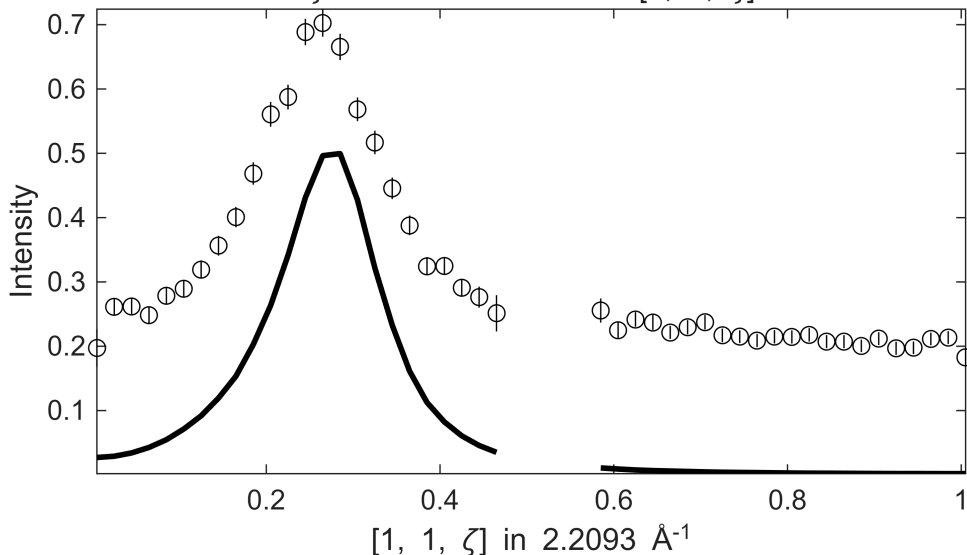


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 226692 pixels -- processing data... -----> retained 188732 pixels

Number of points in input object: 226692
 Fraction of object selected: 100.00% (= 226692 points) ; Access speed: 1.99GB/sec
 Fraction of object retained: 83.25% (= 188732 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 20.3%; Transf. time frac.: 48.0%; Resulting pix preparation: 31.6%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $70 \leq E \leq 90$ eV
 $\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



**** $E_n = 90 \pm 10$

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.65359

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 242131 pixels -- processing data... -----> included 200883 pixels

Number of points in input object: 242131

Fraction of object selected: 100.00% (= 242131 points) ; Access speed: 1.92GB/sec
Fraction of object retained: 82.96% (= 200883 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 37.0%; Transf. time frac.: 59.8%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 242111 pixels -- processing data... -----> retained 200863 pixels

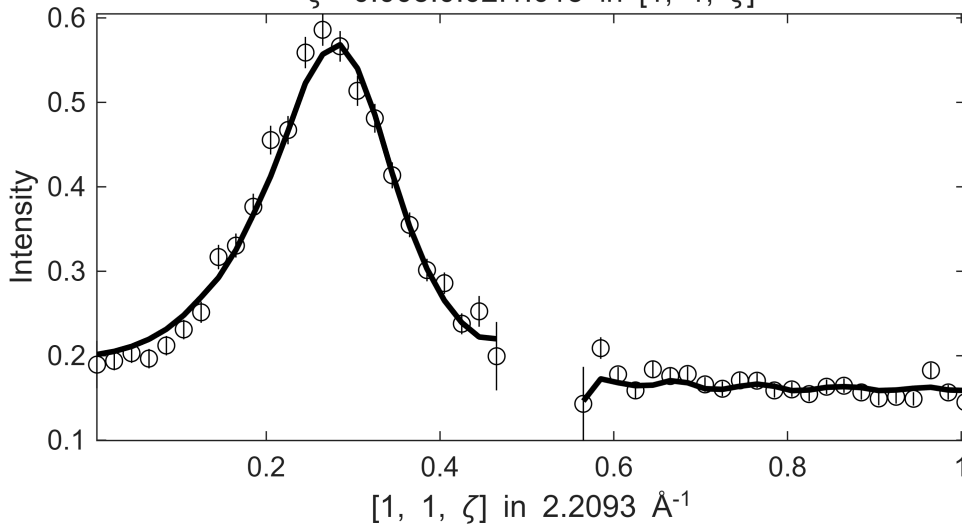
Number of points in input object: 242111

Fraction of object selected: 100.00% (= 242111 points) ; Access speed: 1.45GB/sec
Fraction of object retained: 82.96% (= 200863 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 28.5%; Transf. time frac.: 55.1%; Resulting pix preparation: 16.4%

Fe_ei401_align_sym3.sqw
S= 1.2±0.015; J0= 32±0.24; gamma=53.6782±0.575099
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 80 ≤ E ≤ 1
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 242131 pixels -- processing data... -----> retained 200883 pixels

Number of points in input object: 242131

Fraction of object selected: 100.00% (= 242131 points) ; Access speed: 1.64GB/sec
Fraction of object retained: 82.96% (= 200883 points)

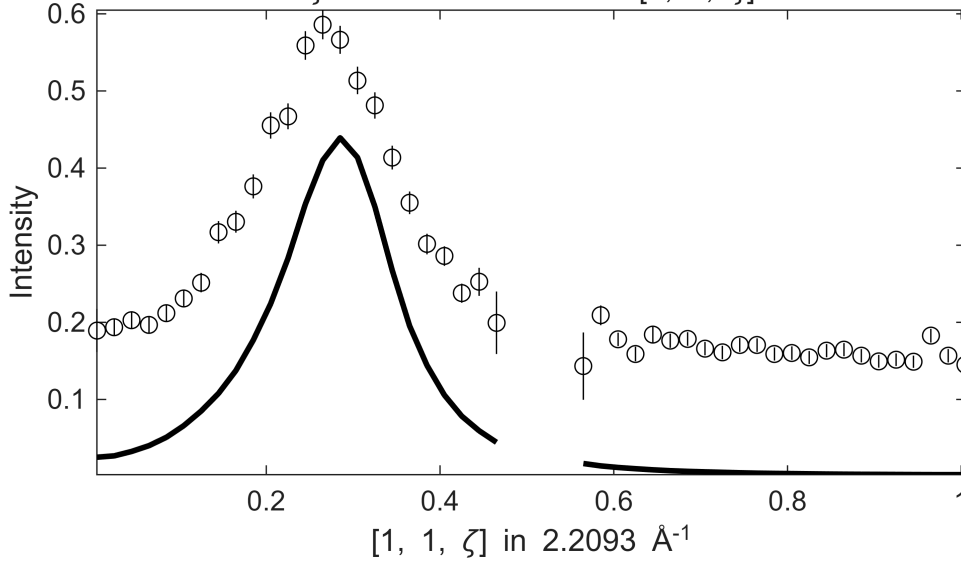
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 27.0%; Transf. time frac.: 53.5%; Resulting pix preparation: 19.5%

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $80 \leq E \leq 1$

$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



**** En = 100±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 267612 pixels -- processing data... -----> included 222408 pixels

Number of points in input object: 267612

Fraction of object selected: 100.00% (= 267612 points) ; Access speed: 1.76GB/sec

Fraction of object retained: 83.11% (= 222408 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 43.0%; Transf. time frac.: 54.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 267609 pixels -- processing data... -----> retained 222405 pixels

Number of points in input object: 267609

Fraction of object selected: 100.00% (= 267609 points) ; Access speed: 1.84GB/sec

Fraction of object retained: 83.11% (= 222405 points)

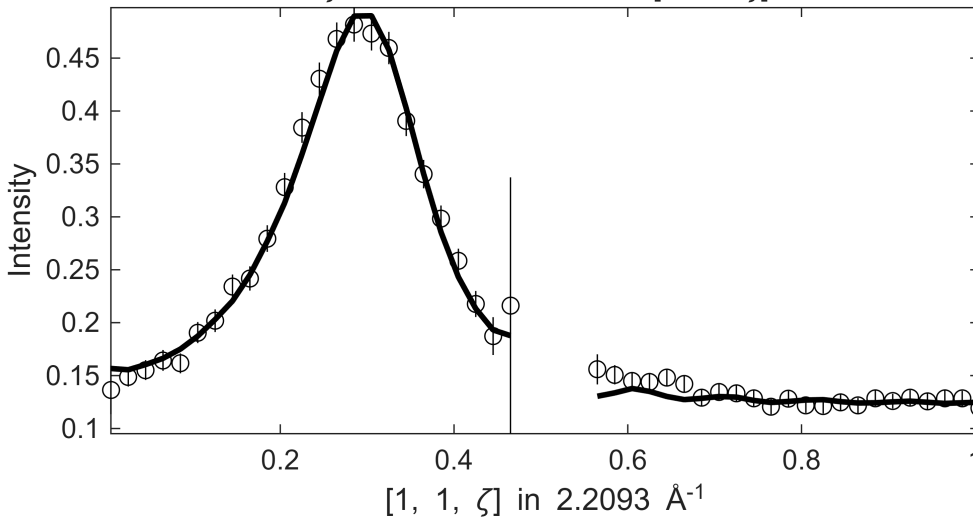
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 24.3%; Transf. time frac.: 57.7%; Resulting pix preparation: 18.0%

Fe_ei401_align_sym3.sqw

S= 1.1±0.014; J0= 32±0.23; gamma=53.8137±0.590958

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 90 ≤ E ≤ 1
ζ=-0.005:0.02:1.015 in [1, 1, ζ]

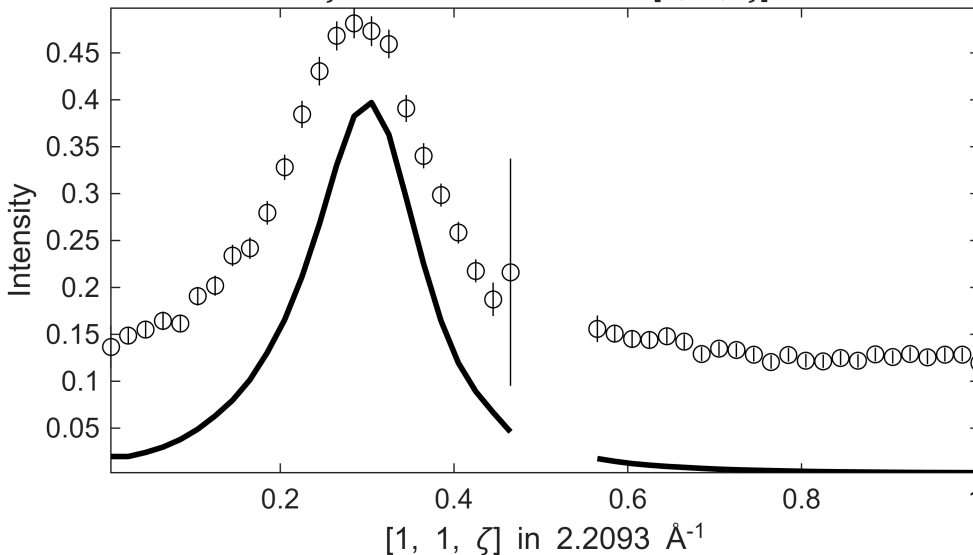


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 267612 pixels -- processing data... -----> retained 222408 pixels
-----
Number of points in input object: 267612
Fraction of object selected: 100.00% (= 267612 points) ; Access speed: 1.57GB/sec
Fraction of object retained: 83.11% (= 222408 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 26.0%; Transf. time frac.: 60.1%; Resulting pix preparation: 13.9%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 90 ≤ E ≤ 1
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



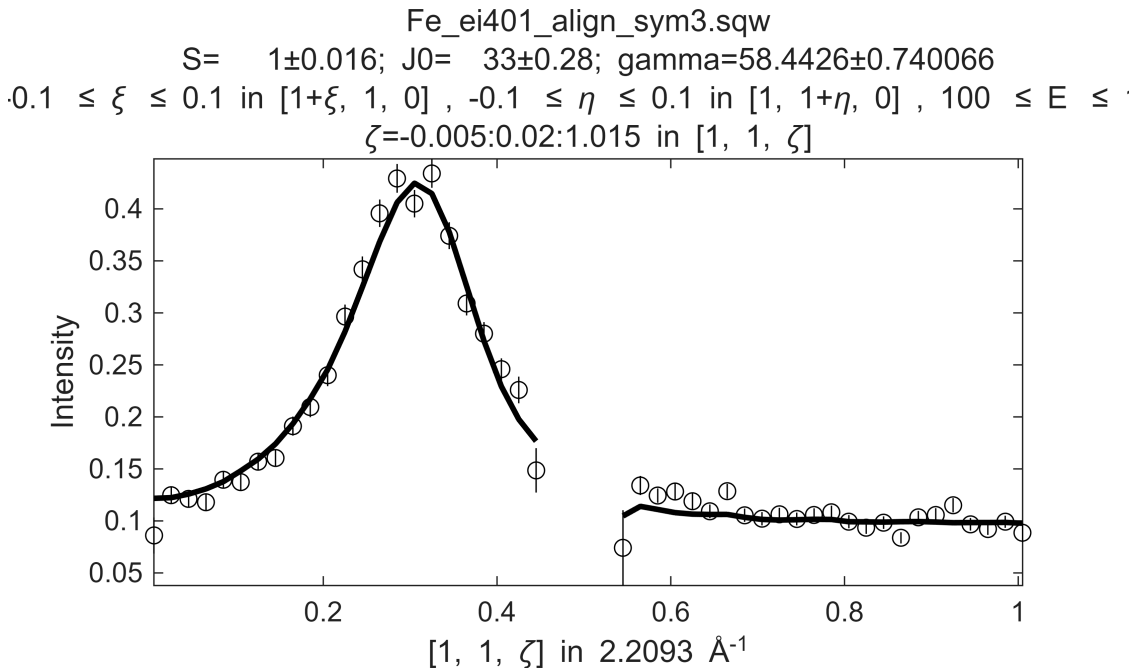
```
*****
**** En = 110±10
*****
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.65359
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 304485 pixels -- processing data... -----> included 253187 pixels
-----
Number of points in input object: 304485
Fraction of object selected: 100.00% (= 304485 points) ; Access speed: 1.60GB/sec
Fraction of object retained: 83.15% (= 253187 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 45.2%; Transf. time frac.: 53.0%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  304460 pixels -- processing data... -----> retained  253164 pixels
-----
Number of points in input object: 304460
      Fraction of object selected:  100.00% (=  304460 points) ; Access speed:      1.60GB/sec
      Fraction of object retained:   83.15% (=  253164 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 31.5%; Transf. time frac.: 40.6%; Resulting pix preparation: 27.9%
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*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  304485 pixels -- processing data... -----> retained  253187 pixels
-----
Number of points in input object: 304485
      Fraction of object selected:  100.00% (=  304485 points) ; Access speed:      1.42GB/sec
      Fraction of object retained:   83.15% (=  253187 points)

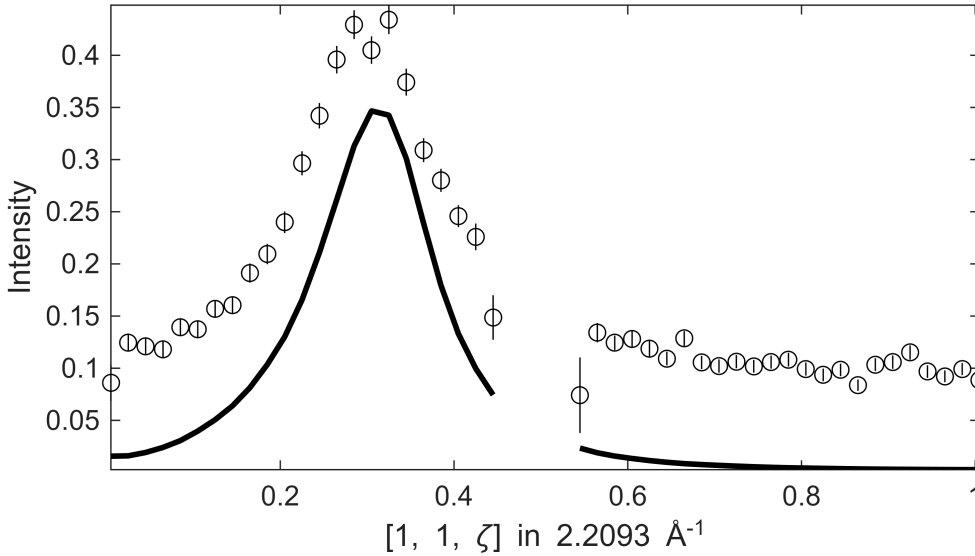
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 32.4%; Transf. time frac.: 48.8%; Resulting pix preparation: 18.8%
-----

```

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $100 \leq E \leq$

$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



**** En = 120±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 342450 pixels -- processing data... -----> included 286588 pixels

Number of points in input object: 342450

Fraction of object selected: 100.00% (= 342450 points) ; Access speed: 1.96GB/sec

Fraction of object retained: 83.69% (= 286588 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 44.6%; Transf. time frac.: 52.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 342448 pixels -- processing data... -----> retained 286586 pixels

Number of points in input object: 342448

Fraction of object selected: 100.00% (= 342448 points) ; Access speed: 1.78GB/sec

Fraction of object retained: 83.69% (= 286586 points)

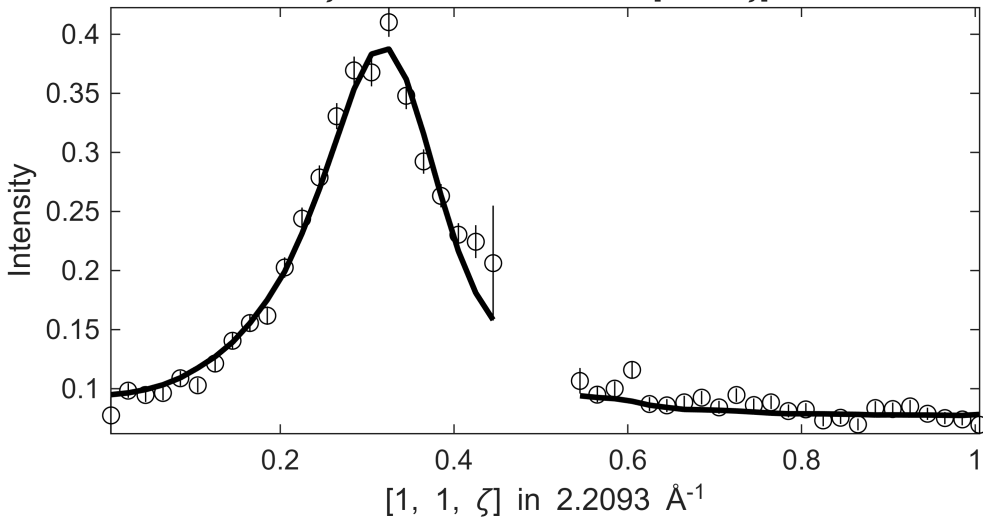
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 29.4%; Transf. time frac.: 54.2%; Resulting pix preparation: 16.3%

Fe_ei401_align_sym3.sqw

S=0.99±0.014; J0= 34±0.26; gamma=58.2209±0.715186

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 110 ≤ E ≤
 ζ=-0.005:0.02:1.015 in [1, 1, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 342450 pixels -- processing data... -----> retained 286588 pixels

Number of points in input object: 342450

Fraction of object selected: 100.00% (= 342450 points) ; Access speed: 1.87GB/sec

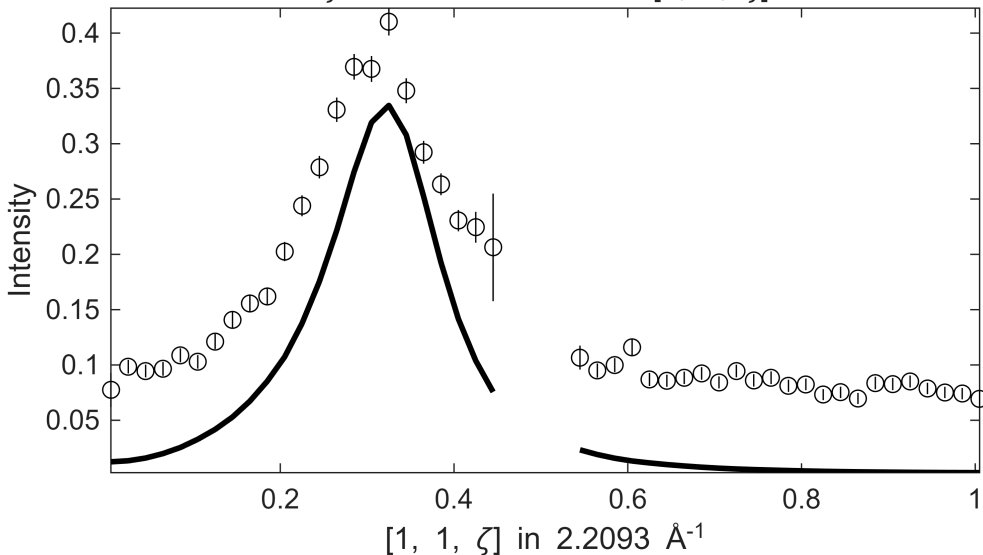
Fraction of object retained: 83.69% (= 286588 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 26.9%; Transf. time frac.: 54.0%; Resulting pix preparation: 19.1%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 110 ≤ E ≤
 ζ=-0.005:0.02:1.015 in [1, 1, ζ]



**** En = 130±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 380465 pixels -- processing data... -----> included 313569 pixels

Number of points in input object: 380465

Fraction of object selected: 100.00% (= 380465 points) ; Access speed: 1.97GB/sec

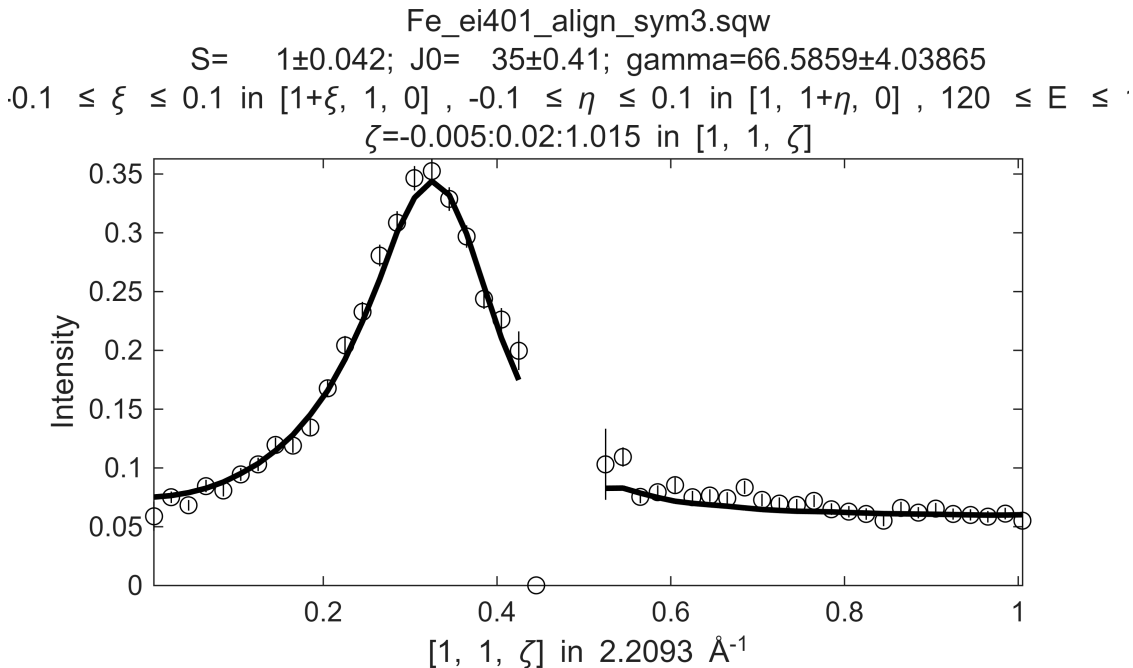
Fraction of object retained: 82.42% (= 313569 points)

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 44.1%; Transf. time frac.: 53.5%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  380442 pixels -- processing data... -----> retained  313567 pixels
-----
Number of points in input object: 380442
      Fraction of object selected:  100.00% (=  380442 points) ; Access speed:      1.90GB/sec
      Fraction of object retained:   82.42% (=  313567 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 29.6%; Transf. time frac.: 51.7%; Resulting pix preparation: 18.6%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  380465 pixels -- processing data... -----> retained  313569 pixels
-----
Number of points in input object: 380465
      Fraction of object selected:  100.00% (=  380465 points) ; Access speed:      1.91GB/sec
      Fraction of object retained:   82.42% (=  313569 points)

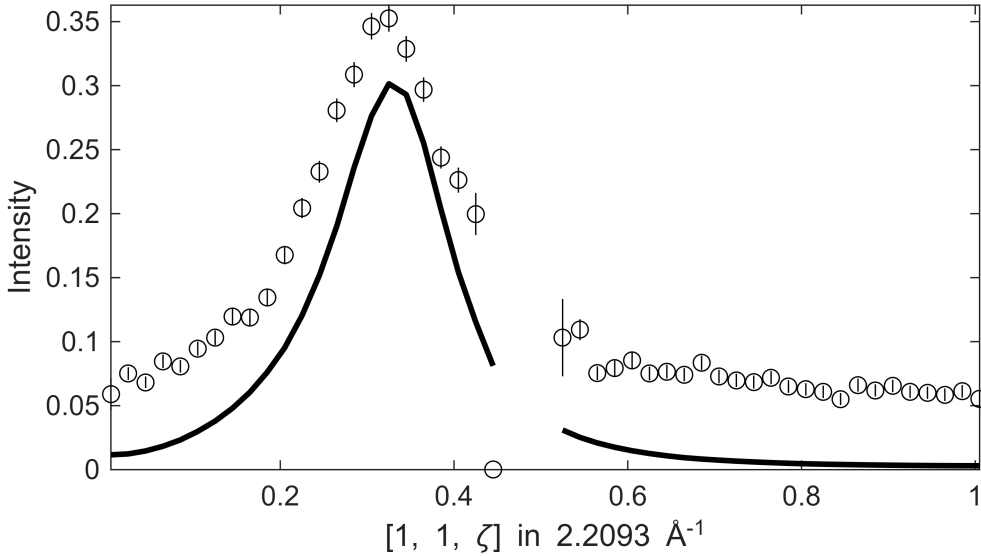
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction: 22.5%; Transf. time frac.: 55.5%; Resulting pix preparation: 22.0%
-----

```

Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $120 \leq E \leq$

$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



**** En = 140±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 491187 pixels -- processing data... -----> included 401328 pixels

Number of points in input object: 491187

Fraction of object selected: 100.00% (= 491187 points) ; Access speed: 1.76GB/sec

Fraction of object retained: 81.71% (= 401328 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 57.2%; Transf. time frac.: 41.1%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 491187 pixels -- processing data... -----> retained 401328 pixels

Number of points in input object: 491187

Fraction of object selected: 100.00% (= 491187 points) ; Access speed: 2.04GB/sec

Fraction of object retained: 81.71% (= 401328 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 32.7%; Transf. time frac.: 45.9%; Resulting pix preparation: 21.3%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 491187 pixels -- processing data... -----> retained 401328 pixels

Number of points in input object: 491187

Fraction of object selected: 100.00% (= 491187 points) ; Access speed: 2.00GB/sec

Fraction of object retained: 81.71% (= 401328 points)

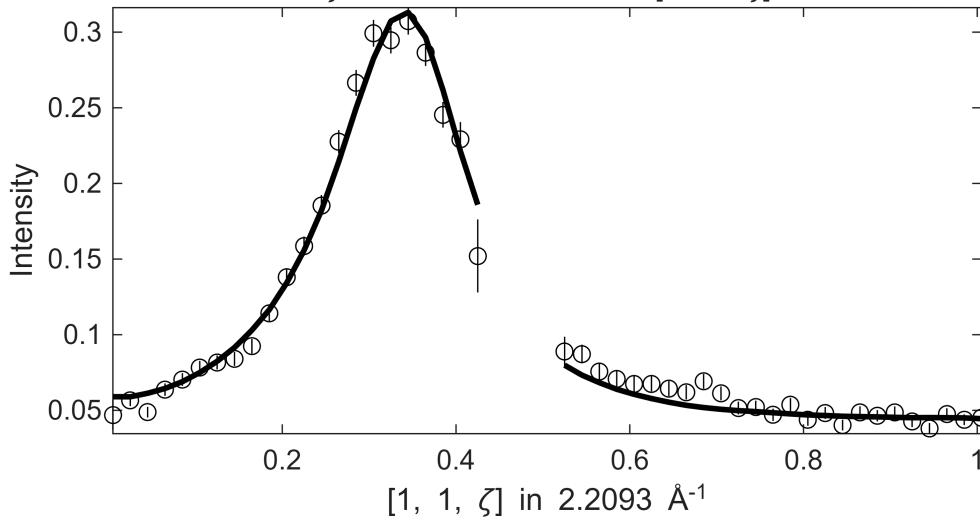
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 30.5%; Transf. time frac.: 46.3%; Resulting pix preparation: 23.2%

Fe_ei401_align_sym3.sqw

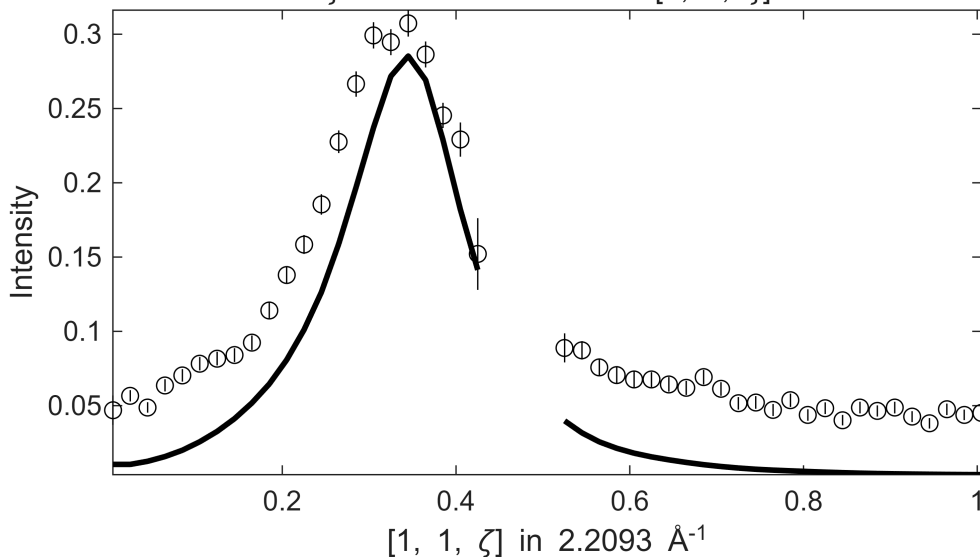
S= 1±0.038; J0= 36±0.36; gamma=71.3906±3.72107

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 130 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 130 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



**** En = 150±10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.3268

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 568777 pixels -- processing data... -----> included 502448 pixels

Number of points in input object: 568777

Fraction of object selected: 100.00% (= 568777 points) ; Access speed: 2.00GB/sec

Fraction of object retained: 88.34% (= 502448 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 54.2%; Transf. time frac.: 44.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; returning pixels - kept

*** Step 1 of 1; Read data for 568579 pixels -- processing data... -----> retained 502250 pixels

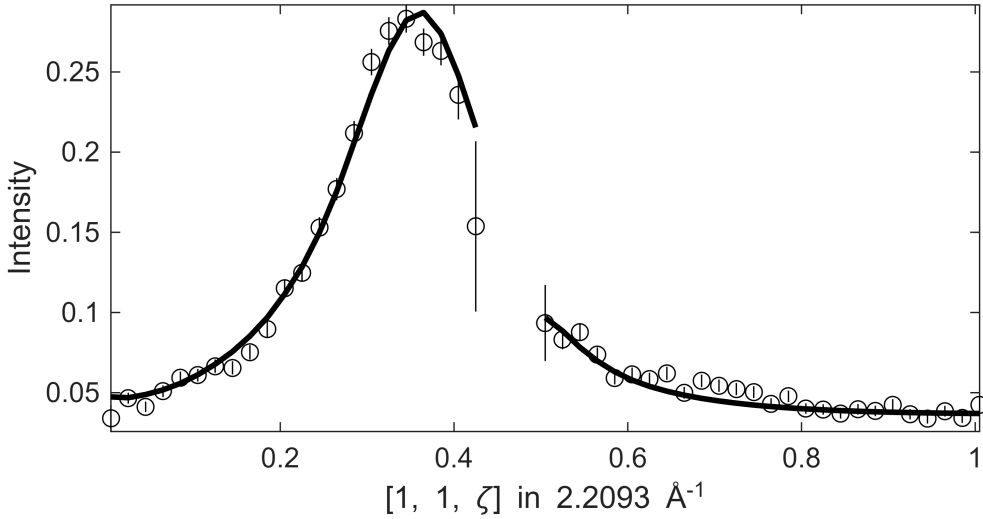
Number of points in input object: 568579

Fraction of object selected: 100.00% (= 568579 points) ; Access speed: 2.08GB/sec

Fraction of object retained: 88.33% (= 502250 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 30.9%; Transf. time frac.: 49.1%; Resulting pix preparation: 19.9%

Fe_ei401_align_sym3.sqw
 $S = 1.1 \pm 0.022$; $J_0 = 35 \pm 0.28$; $\gamma = 82.1173 \pm 1.35432$
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $140 \leq E \leq$
 $\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$

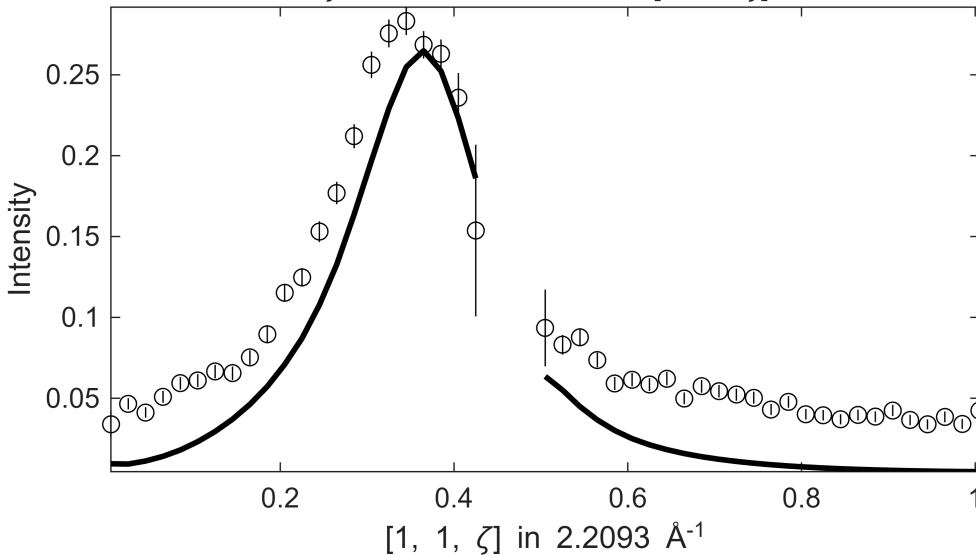


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 568777 pixels -- processing data... -----> retained 502448 pixels

Number of points in input object: 568777
 Fraction of object selected: 100.00% (= 568777 points); Access speed: 2.09GB/sec
 Fraction of object retained: 88.34% (= 502448 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 24.2%; Transf. time frac.: 47.8%; Resulting pix preparation: 27.9%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $140 \leq E \leq$
 $\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



**** $E_n = 160 \pm 10$

Warning: Dataset:24 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.8431

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 399481 pixels -- processing data... -----> included 341153 pixels

Number of points in input object: 399481

Fraction of object selected: 100.00% (= 399481 points) ; Access speed: 2.10GB/sec
Fraction of object retained: 85.40% (= 341153 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 48.6%; Transf. time frac.: 49.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 398827 pixels -- processing data... -----> retained 340516 pixels

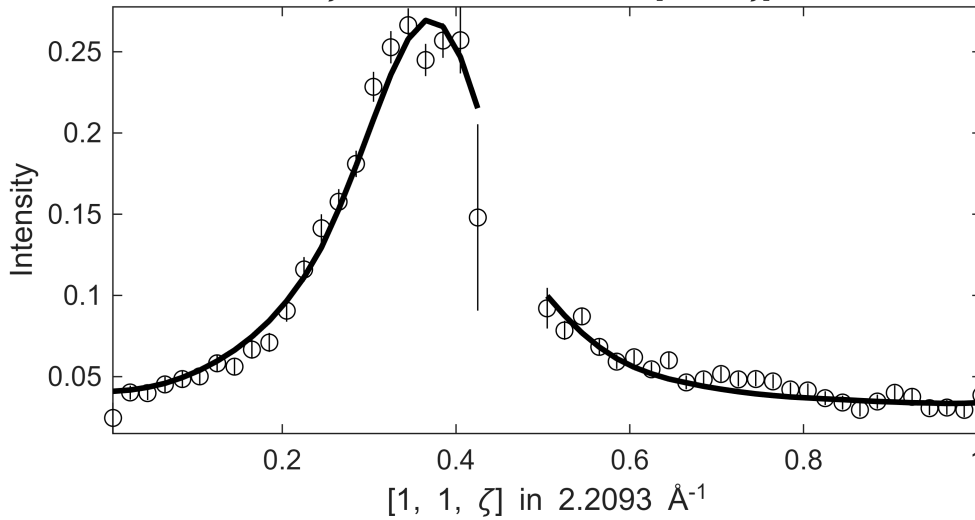
Number of points in input object: 398827

Fraction of object selected: 100.00% (= 398827 points) ; Access speed: 2.06GB/sec
Fraction of object retained: 85.38% (= 340516 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 31.7%; Transf. time frac.: 47.9%; Resulting pix preparation: 20.4%

Fe_ei401_align_sym3.sqw
S= 1.1±0.051; J0= 35±0.57; gamma=82.9207±5.37303
0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 150 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 399481 pixels -- processing data... -----> retained 341153 pixels

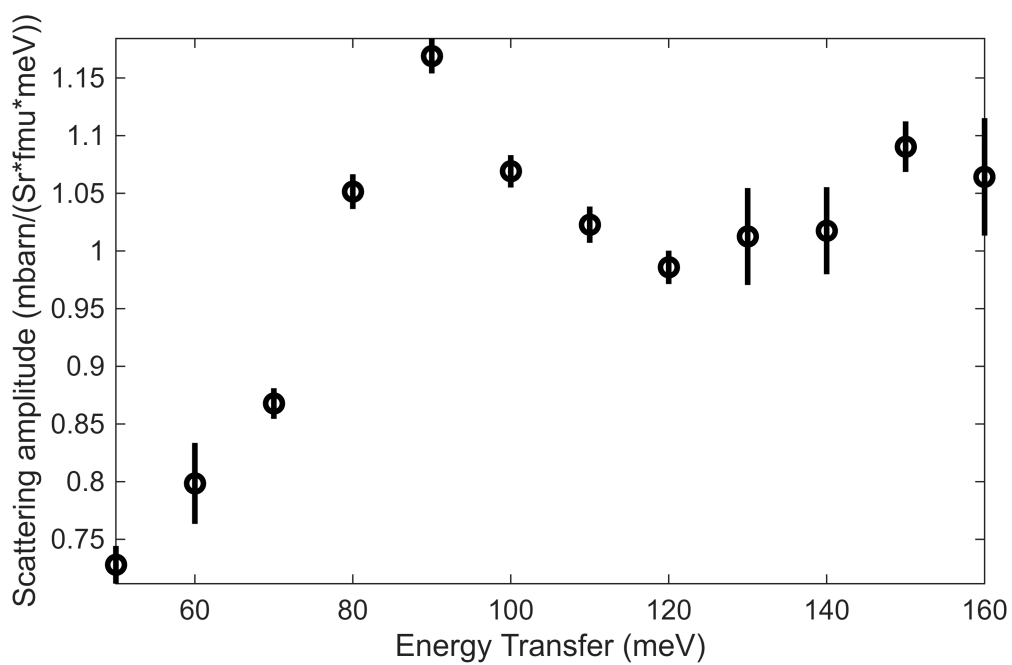
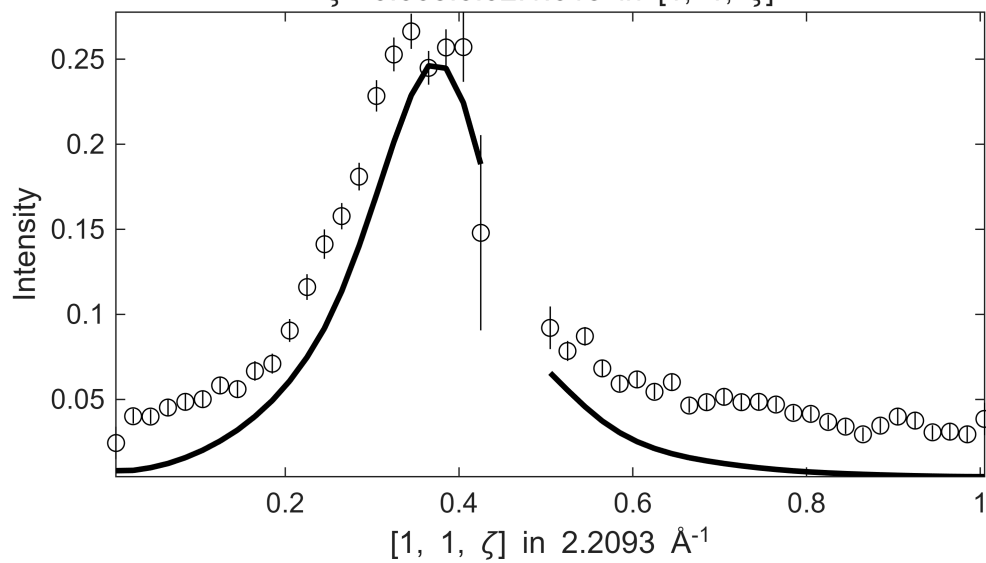
Number of points in input object: 399481

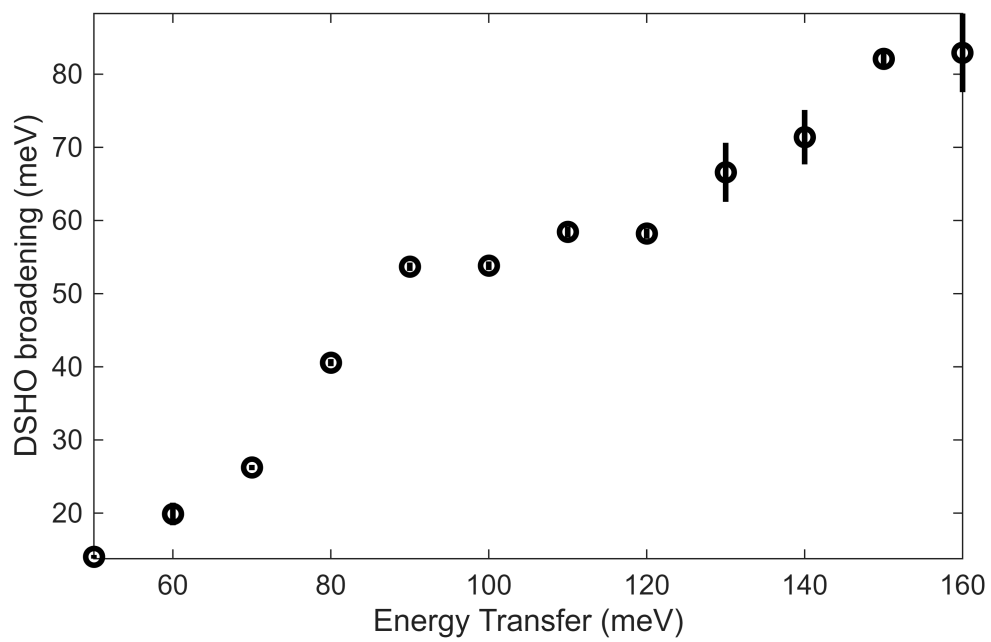
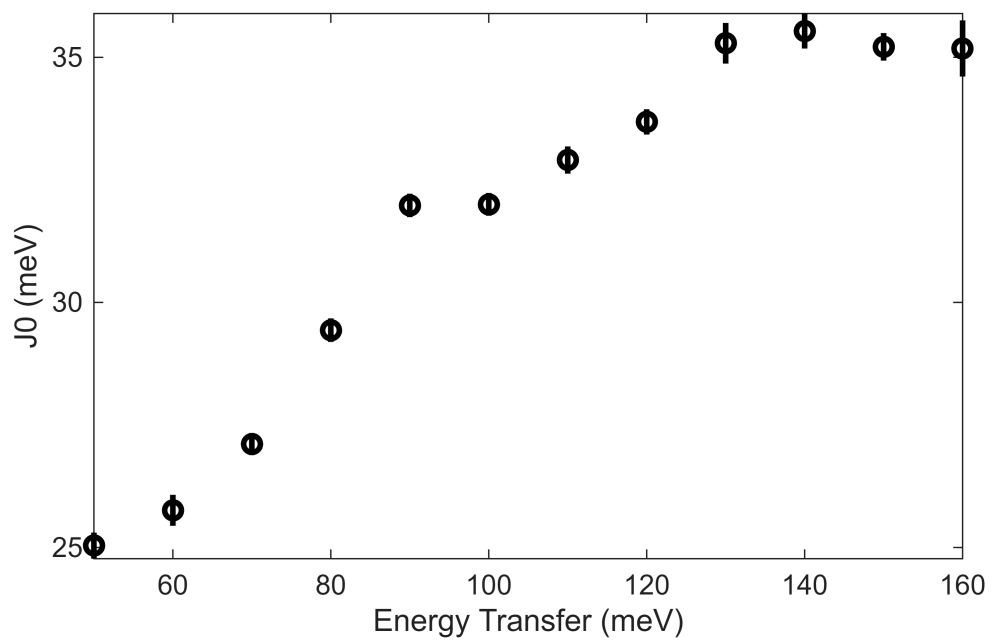
Fraction of object selected: 100.00% (= 399481 points) ; Access speed: 2.03GB/sec
Fraction of object retained: 85.40% (= 341153 points)

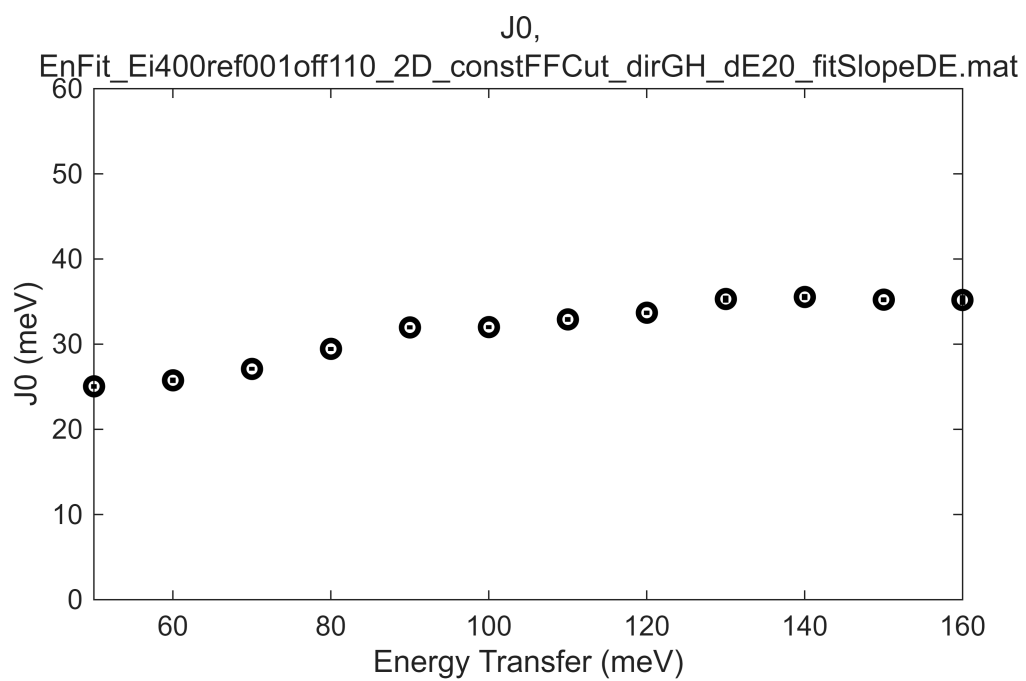
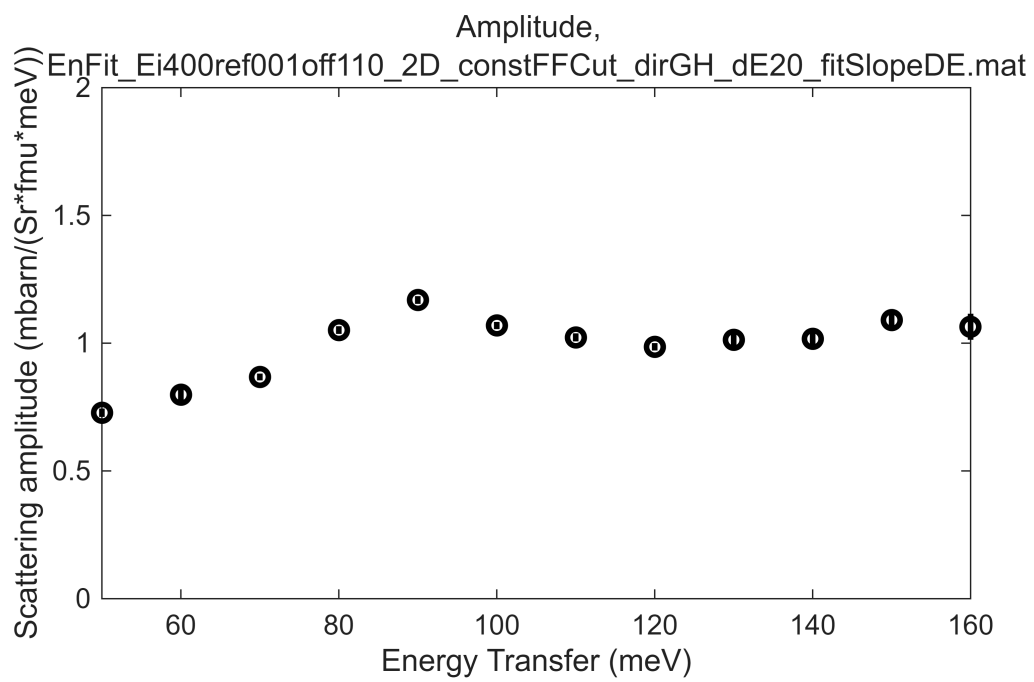
**** Total time in cut(sqw) : 0.1sec Including:

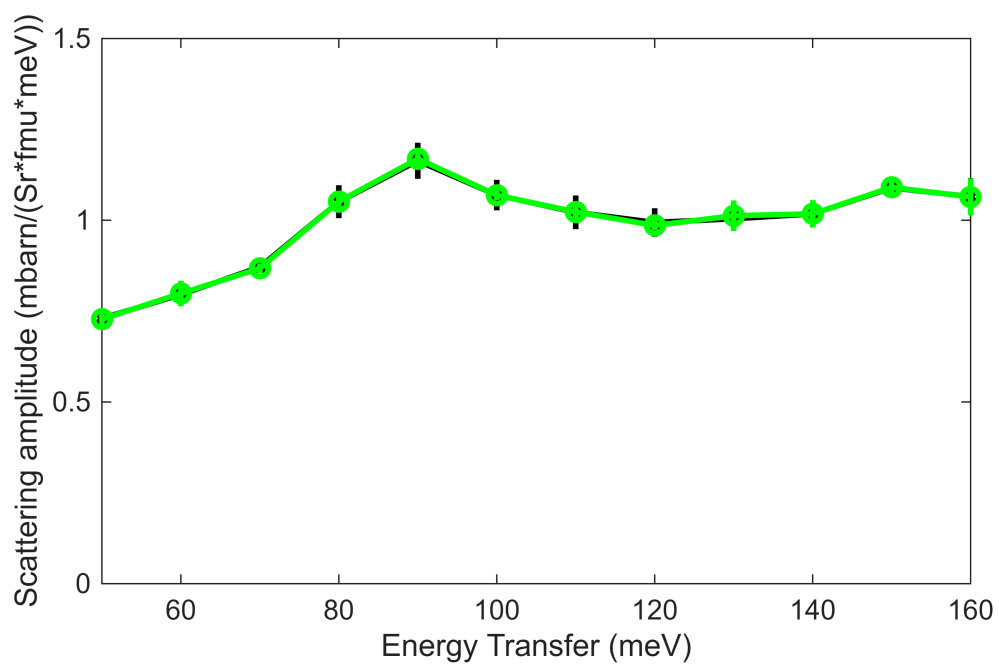
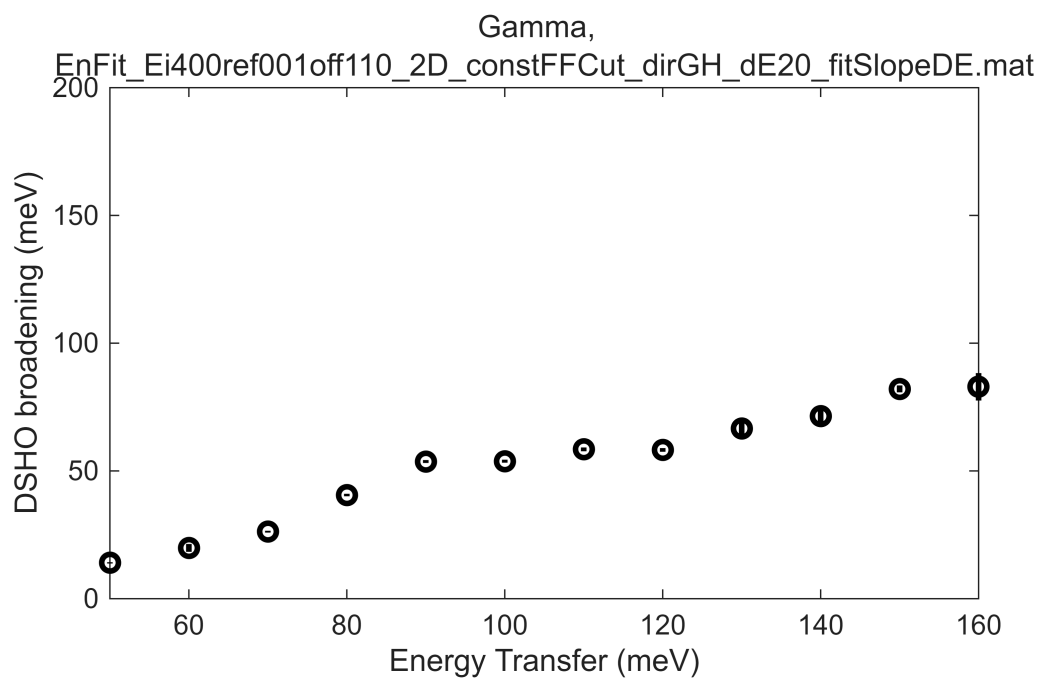
Data access time fraction: 25.4%; Transf. time frac.: 50.2%; Resulting pix preparation: 24.4%

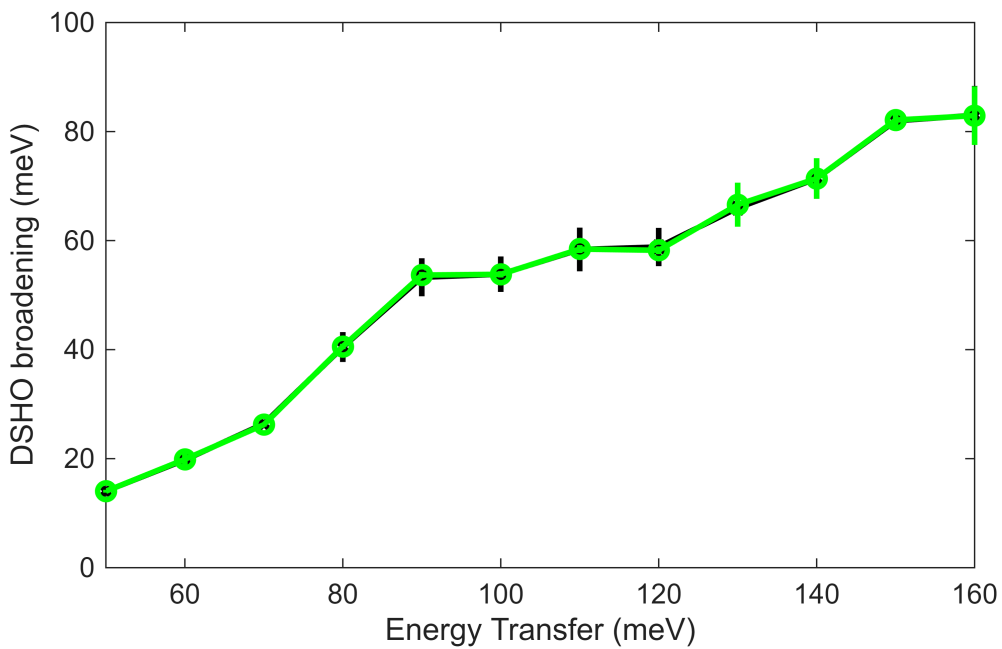
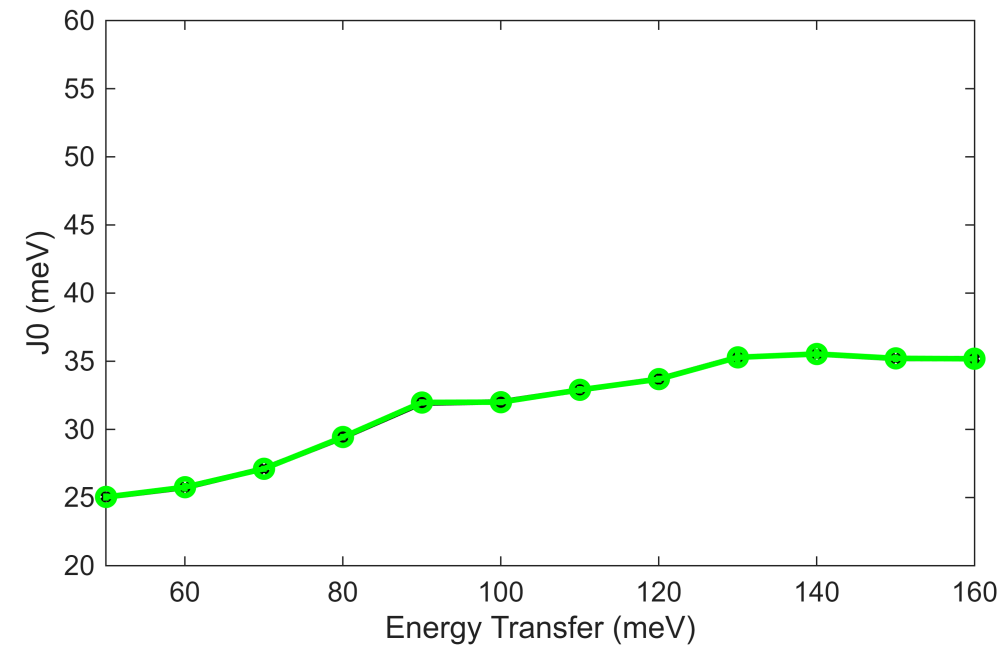
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $150 \leq E \leq 155$ meV
 $\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$







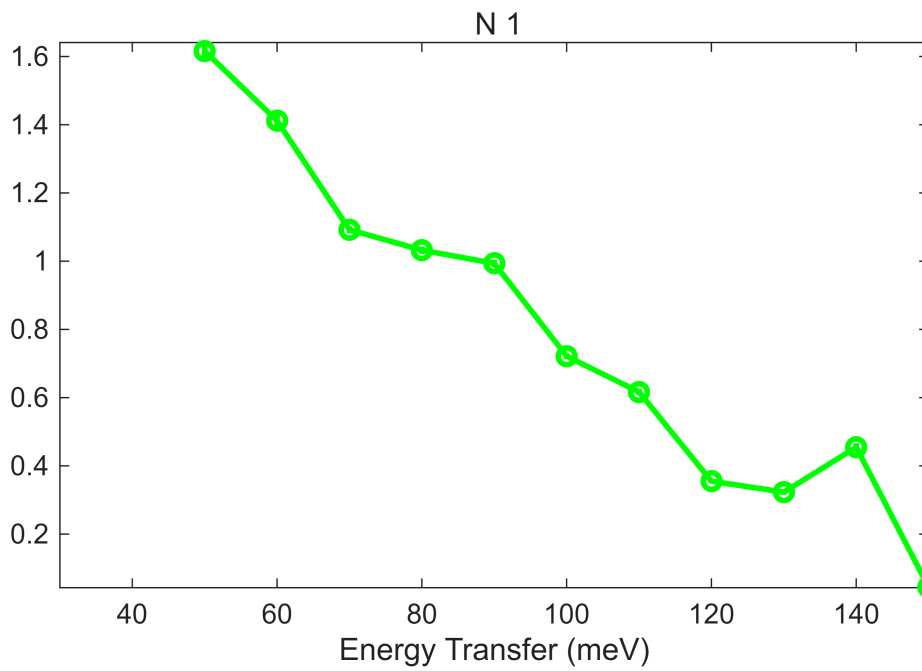




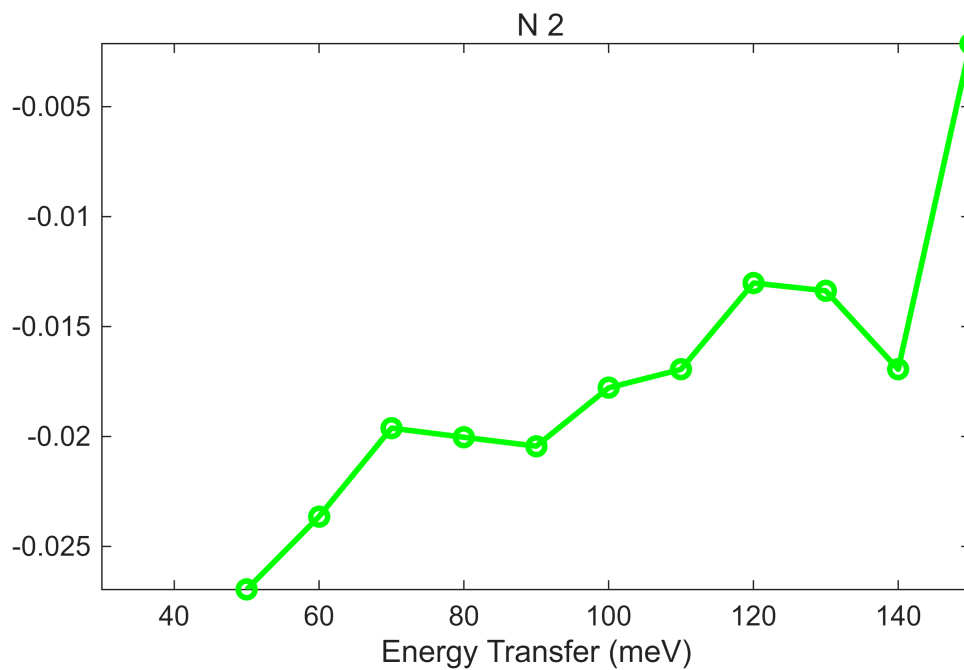
```
%c1,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
beep
beep
beep
```

```
fit_res_400 = fit_res_400rec;

bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;
```

```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure
```



```
fit_resEi400 = fit_res_400rec;
save(res_name,"fit_resEi400")
```